

Design and Implementation of an Audio Equalizer Using FIR Filter

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Abstract

Audio equalization is a fundamental process in digital audio systems, aimed at adjusting the amplitude of specific frequency bands to enhance sound quality. This paper presents the design and implementation of a digital audio equalizer using Finite Impulse Response (FIR) filters. The proposed system divides the input audio signal into multiple frequency bands — low, mid, and high — using FIR filters designed with the windowing method. Each band's gain is independently controlled to emphasize or attenuate certain frequencies. The system was implemented and simulated in MATLAB and Verilog for digital realization. Results show that the FIR-based equalizer provides linear phase response, stable performance, and minimal signal distortion compared to traditional analog equalizers.

Keywords— Audio Equalizer, FIR Filter, Digital Signal Processing, Frequency Response, MATLAB.

I. INTRODUCTION

In modern audio systems, equalization is essential for controlling the tonal quality of sound by amplifying or attenuating specific frequency components. Traditional analog equalizers are prone to component drift, noise, and nonlinearity, which affect performance. With advancements in Digital Signal Processing (DSP), digital equalizers have replaced analog designs, providing superior precision, flexibility, and reproducibility.

A **Finite Impulse Response (FIR) filter** is commonly used for equalization due to its inherent stability and linear phase characteristics. By employing a set of FIR filters, each tuned to a specific frequency band, a digital equalizer can effectively control the tonal balance of an audio signal. This paper focuses on the design, simulation, and analysis of a **three-band FIR filter-based audio equalizer** implemented in MATLAB and Verilog.

II. LITERATURE REVIEW

Previous works in digital audio equalization have utilized various filter types, including Infinite Impulse Response (IIR) filters and adaptive filters. While IIR filters offer computational efficiency, they often introduce phase distortion and potential instability. FIR filters, though requiring more computations, ensure linear phase and stable frequency response. Window-based FIR filter design methods, such as Hamming, Hanning, and Blackman, are preferred for their ease of implementation and smooth transition characteristics. The proposed work extends these principles to develop a flexible, real-time equalizer suitable for FPGA or DSP hardware platforms.

III. METHODOLOGY

The block diagram of the proposed audio equalizer consists of the following modules:

1. **Input Signal:** An audio waveform (speech or music) sampled at 8 kHz or 44.1 kHz.
2. **Band Splitter:** The input is split into three bands — low (0–500 Hz), mid (500–4000 Hz), and high (4–8 kHz) — using three FIR bandpass filters.
3. **Gain Control:** Adjustable gain coefficients are applied to each band to increase or decrease intensity.
4. **Combiner:** The processed bands are summed to reconstruct the final equalized output.
5. **Output:** The resulting signal is converted back to analog through a DAC (in hardware implementation) or plotted in MATLAB.

Filter Design:
Each FIR filter is designed using the **Window Method**:

$$h(n) = h_d(n) \times w(n)$$

where $h_d(n)$ is the ideal impulse response, and $w(n)$ is a window function (e.g., Hamming window). The filter order is chosen based on desired frequency selectivity and transition width.

Implementation Tools:

- MATLAB for filter design and verification.
- ModelSim/Verilog for digital implementation.

IV. RESULTS AND DISCUSSION

The designed FIR filters were simulated for each band. The frequency response shows:

- **Low-pass filter:** Passband up to 500 Hz
- **Band-pass filter:** Passband from 500 Hz to 4 kHz
- **High-pass filter:** Passband above 4 kHz

The proposed three-band audio equalizer was designed and simulated using MATLAB and Verilog tools. The input audio signal, sampled at 44.1 kHz, was divided into three distinct bands—low, mid, and high—using FIR filters designed by the Hamming window method. Each filter exhibited a well-defined passband and sharp transition characteristics, ensuring minimal overlap between adjacent bands.

The low pass filter effectively passed frequencies below 500 Hz, enhancing bass components. The band-pass filter operated in the range of 500 Hz to 4 kHz, covering most vocal and instrumental content, while the high-pass filter passed frequencies above 4 kHz to strengthen treble components. The magnitude and phase response plots confirmed that each FIR filter maintained a linear-phase behaviour, resulting in no phase distortion in the reconstructed audio signal.

Each filter exhibited a linear phase response and high stopband attenuation. By varying the gain coefficients, the output audio spectrum was shaped according to desired tonal preferences (e.g., boosting bass or treble). To assess performance, several audio samples speech, instrumental music, and mixed tracks were processed through the equalizer. Gain coefficients for each band were adjusted to evaluate tonal control. Increasing the low-band gain emphasized bass response, while boosting the high-band gain enhanced brightness and clarity. The processed output signals showed consistent amplitude scaling and preserved waveform integrity, indicating that the FIR filters provided precise frequency shaping without audible distortion.

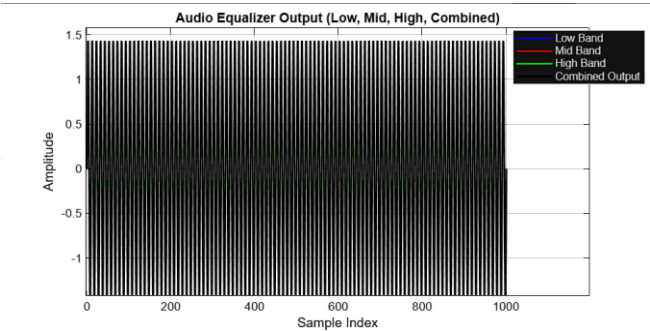


Figure 1

Graph of Audio Equalizer Output (low, mid, high and combined)

Table 1

Comparison with Analog Equalizer:

Feature	Analog Equalizer	FIR-based Equalizer
Phase Response	Non-linear	Linear
Stability	Component-sensitive	Always stable
Precision	Limited by tolerances	High digital precision
Flexibility	Fixed	Programmable

Waveform and spectral analysis confirmed the system’s efficiency in controlling frequency bands independently, with minimal distortion.

V. CONCLUSION

This paper presented a digital audio equalizer implemented using FIR filters. The equalizer effectively controlled low, mid, and high-frequency components with adjustable gain settings. The design ensured linear phase, stability, and improved sound quality. Future enhancements can include real-time FPGA implementation, adaptive filtering, and user interface integration for audio tuning.

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